Tutorial 2: FINA2205 Probability and Finance

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Mar 2026

Agenda

- Tutorial 2 (10 problems)

Recap: return and risk

- Historical returns: each observation has equal weight, and returns are usually modelled as a **continuous random variable**.

$$\bar{X} = \frac{1}{n}\sum X_i \qquad S = \sqrt{\frac{\sum(X_i - \bar{X})^2}{n-1}}$$

- Scenario analysis: In finance, however, future returns are often driven by **uncertainty** and different economic scenarios such as recession, normal conditions, and boom.
- In that case, X becomes a **discrete random variable** with a finite set of possible outcomes and **probabilities**.

Recap: return and risk

- Therefore, we move from equal weighting to **probability weighting**.
- Use:

$$E(X) = \sum_{i=1}^{N} X_i P(X = X_i) \quad \sigma = \sqrt{\sum (X_i - E(X))^2 P(X_i)}$$

- Average return $(\bar{X})$ summarizes what happened
- Expected return $(E(X))$ summarizes what could happen

Recap: return and risk

TABLE 10.1 Probability Distribution of Returns for BFI

Current Stock Price ($)	Stock Price in One Year ($)	Probability Distribution: Return, R	Probability Distribution: Probability, P_R	
	140	0.40	25%	boom
100	110	0.10	50%	normal
	80	−0.20	25%	recession

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

$$E[R_{BFI}] = 25\%(-0.20) + 50\%(0.10) + 25\%(0.40) = 10\%$$

Variance: The expected squared deviation from the mean

$$Var(R) = E\left[\left(R - E[R]\right)^2\right] = \sum_R P_R \times \left(R - E[R]\right)^2$$

$$Var[R_{BFI}] = 25\% \times (-0.20 - 0.10)^2 + 50\% \times (0.10 - 0.10)^2 + 25\% \times (0.40 - 0.10)^2 = 0.045$$

Standard Deviation: square root of variance

$$SD(R) = \sqrt{Var(R)}$$

$$SD(R) = \sqrt{Var(R)} = \sqrt{0.045} = 21.2\%$$

Recap: beyond return

- Scenario returns introduced a discrete random variable.
- More broadly, a discrete random variable has a **countable** set of possible values.
- Common **naturally** discrete examples:
  - number of classes (0, 1, 2, 3....); number of bankruptcies
- So the **same** probability-weighted formulas apply to naturally discrete variables as well.

2-1

1. The number of arrivals per minute at a bank located in the central business district of a large city was recorded over a period of 200 minutes with the results shown in the table below. Complete (a) and (b) to the right.

Arrivals	Frequency
0	12
1	28
2	48
3	40
4	29
5	22
6	11
7	7
8	3

a. Compute the expected number of arrivals per minute.

$\mu =$ _____

(Type an integer or decimal rounded to three decimal places as needed.)

b. Compute the standard deviation.

$\sigma =$ _____

(Type an integer or decimal rounded to three decimal places as needed.)

2-1 solution

a.

$$E(X) = \sum_{i=1}^{N} X_i P(X = X_i)$$

- The expected number of arrivals per minute is 3.045
- $\left(0\times\frac{12}{200} + 1\times\frac{28}{200} + 2\times\frac{48}{200} + 3\times\frac{40}{200} + 4\times\frac{29}{200} + 5\times\frac{22}{200} + 6\times\frac{11}{200} + 7\times\frac{7}{200} + 8\times\frac{3}{200}\right) = 3.045$

2-1 solution (cont.)

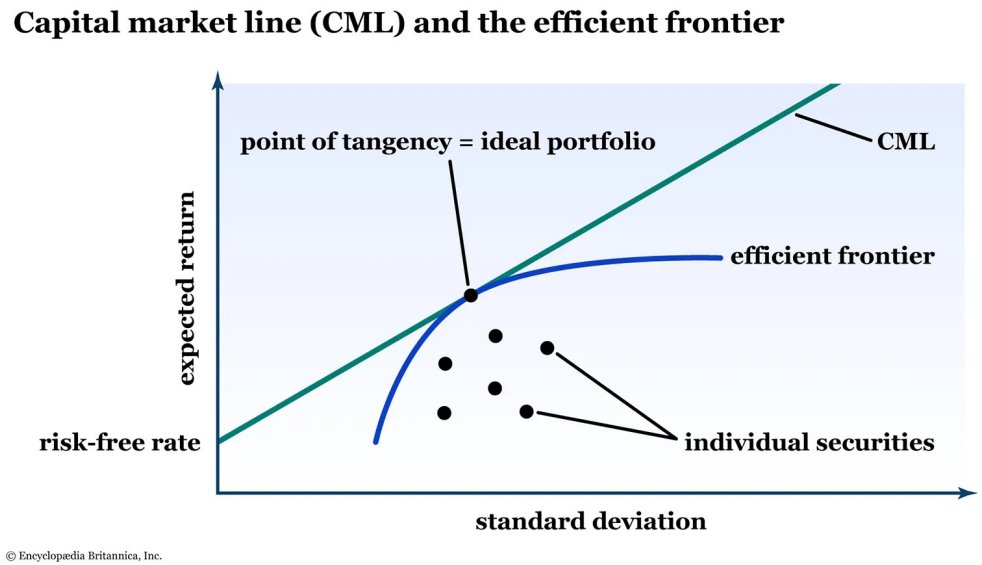
b.

$$\sigma = \sqrt{\sum (X_i - E(X))^2 P(X_i)}$$

- Standard deviation: 1.831
- $\left[(0-3.045)^2 \times \frac{12}{200} + (1-3.045)^2 \times \frac{28}{200} + (2-3.045)^2 \times \frac{48}{200} + (3-3.045)^2 \times \frac{40}{200} + (4-3.045)^2 \times \frac{29}{200} + (5-3.045)^2 \times \frac{22}{200} + (6-3.045)^2 \times \frac{11}{200} + (7-3.045)^2 \times \frac{7}{200} + (8-3.045)^2 \times \frac{3}{200}\right]^{1/2} = 1.831$

Recap: portfolio

- For a single security, risk is captured by return volatility.
- For a portfolio, risk is driven by both individual volatilities and **co-movement** across assets.
- Markowitz (1952) showed that diversification works because imperfect correlation can reduce portfolio risk.



Modern portfolio theory (MPT)

Recap: portfolio

- In a scenario-based setting, the **covariance** (σ_{XY}) is:

$$Cov(X,Y) = \sigma_{XY} = \sum (X_i - E(X))(Y_i - E(Y)P_i$$

- The **coefficient of correlation** (correlation coefficient, ρ_{XY}) standardizes covariance:

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

where $-1 < \rho_{XY} < 1$.

- Interpretation: (1) The closer to -1, the stronger the negative linear relationship; (2) The closer to 1, the positive the negative linear relationship; (3) The closer to 0, the weaker the linear relationship.

Recap: portfolio

A portfolio of two securities

- Let x_1 be the weight invested in Security X, and x_2 the weight invested in Security Y.
- The portfolio expected return is:

$$E(P) = x_1 E(X) + x_2 E(Y)$$

- The portfolio risk (σ_P) is:

$$\sigma_P = \sqrt{x_1^2\sigma_1^2 + 2x_1x_2\sigma_X\sigma_Y\rho_{XY} + x_2^2\sigma_2^2}$$

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y} \rightarrow \sigma_{XY} = \sigma_X\sigma_Y\rho_{XY} \rightarrow \sigma_P = \sqrt{x_1^2\sigma_1^2 + 2x_1x_2\sigma_{XY} + x_2^2\sigma_2^2}$$

$$\sigma_P \neq \sigma_{XY}$$

2-3

3. You are trying to develop a strategy for investing in two different stocks. The anticipated annual return for a $1,000 investment in each stock under four different economic conditions has the probability distribution shown to the right. Complete parts (a) through (d).

		Returns	
Probability	Economic Condition	Stock X	Stock Y
0.3	Recession	−45	−100
0.1	Slow growth	20	50
0.2	Moderate growth	110	130
0.4	Fast growth	150	225

a. Compute the expected return for stock X and for stock Y.

The expected return for stock X is $_____.
(Type an integer or a decimal.)

The expected return for stock Y is $_____.
(Type an integer or a decimal.)

b. Compute the standard deviation for stock X and for stock Y.

The standard deviation for stock X is $_____.
(Round to two decimal places as needed.)

The standard deviation for stock Y is $_____.
(Round to two decimal places as needed.)

2-3 solution

a.

$$E(X) = \sum_{i=1}^{N} X_i P(X = X_i)$$

- The expected return for stock X is \$70.5
- $E(X) = (-45)\times 0.3 + 20\times 0.1 + 110\times 0.2 + 150\times 0.4 = 70.5$
- The expected return for stock Y is \$91.0
- $E(Y) = (-100)\times 0.3 + 50\times 0.1 + 130\times 0.2 + 225\times 0.4 = 91.0$

2-3 solution

b.

$$\sigma = \sqrt{\sum (X_i - E(X))^2 P(X_i)}$$

- The standard deviation for stock X is $84.25
- $\sigma_X = [(-45 - 70.5)^2 \times 0.3 + (20 - 70.5)^2 \times 0.1 + (110 - 70.5)^2 \times 0.2 + (150 - 70.5)^2 \times 0.4]^{1/2} = 84.25$
- The standard deviation for stock Y is $136.378
- $\sigma_Y = [(-100 - 91)^2 \times 0.3 + (50 - 91)^2 \times 0.1 + (130 - 91)^2 \times 0.2 + (225 - 91)^2 \times 0.4]^{1/2} = 136.378$

2-3 (cont.)

c. Compute the covariance of stock X and stock Y.

The covariance of stock X and stock Y is _____.
(Type an integer or a decimal.)

d. Would you invest in stock X or stock Y? Explain.

Risk-averse investors would invest in stock (1) _____ because it has a (2) _____ whereas risk takers would invest in stock (3) _____ because it has a (4) _____

(1)
- ○ Y
- ○ X

(2)
- ○ lower standard deviation,
- ○ higher expected return,
- ○ lower expected return,
- ○ higher standard deviation,

(3)
- ○ Y
- ○ X

(4)
- ○ higher standard deviation.
- ○ higher expected return.
- ○ lower standard deviation.
- ○ lower expected return.

2-3 solution (cont.)

c.

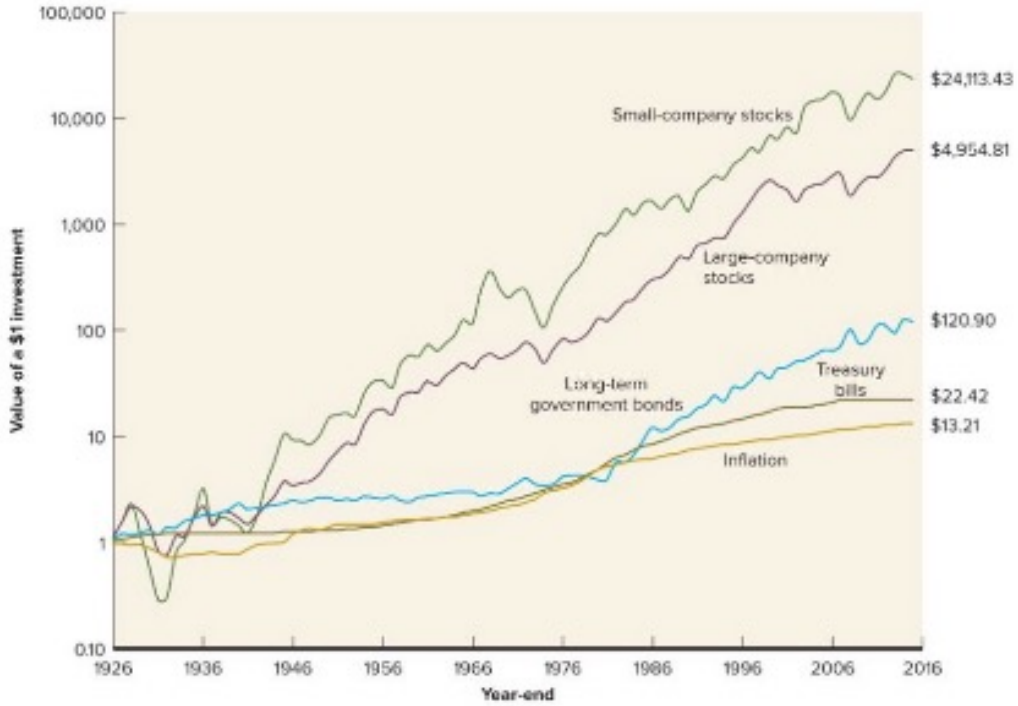
$$\sigma_{XY} = \sum (X_i - E(X))(Y_i - E(Y)P_i$$

- The covariance of stock X and stock Y is 11394.5
- $\sigma_{XY} = (-45 - 70.5)\times(-100 - 91)\times 0.3 + (20 - 70.5)\times(50 - 91)\times 0.1 + (110 - 70.5)\times(130 - 91)\times 0.2 + (150 - 70.5)\times(225 - 91)\times 0.4 = 11394.5$

d.

- Risk-averse investors would invest in stock **X** because it has a **lower standard deviation** whereas risk takers would invest in stock **Y** because it has a **higher standard deviation** and higher return

FIGURE 1.1 A $1 Investment in Different Types of Portfolios: 1926–2015 (Year-end 1925 = $1)



Over 90 years, higher risk received a higher return

2-2

1: Summary of Investment Returns

The table summarizes the estimate of the returns per $1,000 investment under three economic conditions.

$P(X_iY_i)$	Economic Condition	Investment: Dow Jones Fund	Investment: Weak-Economy Fund
0.2	Recession	− $300	+ $200
0.5	Stable economy	+ 200	+ 50
0.3	Expanding economy	+ 250	− 200

2. Suppose that you are deciding between two different investments for the coming year. The first investment is a mutual fund that consists of the stocks that comprise the Dow Jones Industrial Average. The second investment is a mutual fund that is expected to perform best when the economy is weak. Let X be the variable that corresponds to the Dow Jones fund and Y be the variable that corresponds to the weak-economy fund. The expected values and the variances for each variable and the covariance are calculated and the results are shown below.

$E(X) = \$115$, $E(Y) = \$5$, $\sigma_X^2 = 43{,}525$, $\sigma_Y^2 = 21{,}225$, $\sigma_{XY} = -22{,}575$

Scenario 1: 50/50

When half of the portfolio assets are invested in the Dow Jones fund and half in a weak-economy fund, the portfolio expected return is found to be $60 and the portfolio risk is found to be $70.00. Complete parts (a) through (c) below.

[1] Click the icon to view the table that summarizes the estimate of returns.

a. Recalculate the portfolio expected return and the portfolio risk if 30% of the portfolio assets are invested in the Dow Jones fund and 70% in a weak-economy fund.

Scenario 2: 30/70

The portfolio expected return is E(P) = $_____. (Type an integer or a decimal.)

The portfolio risk is σ_P = $_____. (Round to two decimal places as needed.)

b. Recalculate the portfolio expected return and the portfolio risk if 70% of the portfolio assets are invested in the Dow Jones fund and 30% in a weak-economy fund.

Scenario 3: 70/30

The portfolio expected return is E(P) = $_____. (Type an integer or a decimal.)

The portfolio risk is σ_P = $_____. (Round to two decimal places as needed.)

2-2 solution

a.

$$E(P) = x_1 E(X) + x_2 E(Y)$$

- The portfolio expected return is E(P)= $38
- $(0.3\times115 + 0.7\times5) = 38$

$$\sigma_P = \sqrt{x_1^2\sigma_X^2 + 2x_1x_2\sigma_X\sigma_X\rho_{XY} + x_2^2\sigma_Y^2} = \sqrt{x_1^2\sigma_X^2 + 2x_1x_2\boldsymbol{\sigma_{XY}} + x_2^2\sigma_Y^2}$$

- The portfolio risk is 69.541
- $(0.3^2\times43{,}525 + 2\times0.3\times0.7\times(-22{,}575) + 0.7^2\times21{,}225)^{1/2} = 69.541$

2-2 solution (cont.)

b.

$$E(P) = x_1 E(X) + x_2 E(Y)$$

- The portfolio expected return is E(P)= $82
- $(0.7\times115 + 0.3\times5) = 82$

$$\sigma_P = \sqrt{x_1^2\sigma_1^2 + 2x_1x_2\sigma_1\sigma_2\beta_{12} + x_2^2\sigma_2^2} = \sqrt{x_1^2\sigma_1^2 + 2x_1x_2\boldsymbol{\sigma_{12}} + x_2^2\sigma_2^2}$$

- The portfolio risk is 117.286
- $(0.7^2\times43{,}525 + 2\times0.3\times0.7\times(-22{,}575) + 0.3^2\times21{,}225)^{1/2} = 117.286$

2-2 (cont.)

c. Based on the risk-to-return ratio, which of the three investment strategies (30%, 50%, or 70% in the Dow Jones fund) would you recommend? Why?

Investing (1) _____ in the Dow Jones index fund is recommended because it will yield the (2) _____ risk per unit average return.

1: Summary of Investment Returns

The table summarizes the estimate of the returns per $1,000 investment under three economic conditions.

$P(X_iY_i)$	Economic Condition	Investment: Dow Jones Fund	Investment: Weak-Economy Fund
0.2	Recession	− $300	+ $200
0.5	Stable economy	+ 200	+ 50
0.3	Expanding economy	+ 250	− 200

(1) ○ 30%
○ 50%
○ 70%

(2) ○ lowest
○ highest

2-2 solution (cont.)

c.

- Investing **50%** in the Dow Jones index fund is recommended because it will yield the **lowest** risk per unit average return.

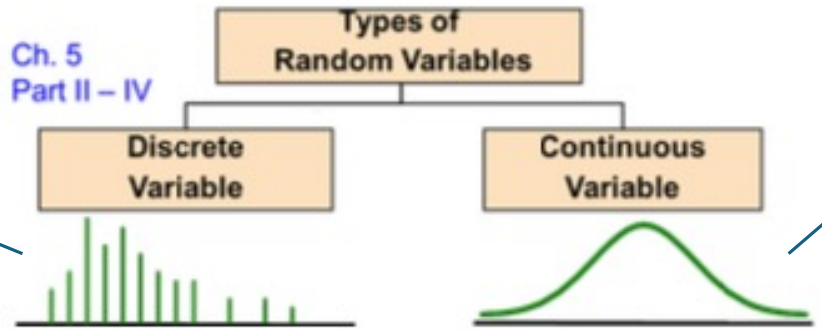
$$\mathrm{R}isk - to - return\ Ratio = \frac{\sigma_P}{E(P)}$$

- Rule: the lower the better.
- $\frac{69.541}{38} = 1.83 > \frac{117.286}{82} = 1.43 > \frac{\mathbf{70}}{\mathbf{60}} = \mathbf{1.17}$

Recap: probability distribution

- How do we describe the probability distribution of X?
- PDF (Probability Density Function) vs CDF (Cumulative Distribution Function)
- PDF describes how probability is distributed across values.
- CDF describes how probability accumulates up to a given value.

PDF

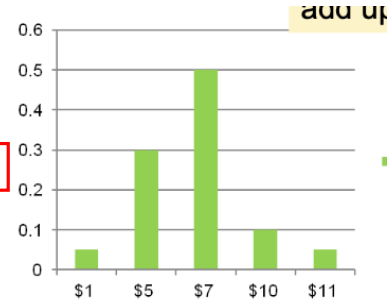


CDF (more meaningful)

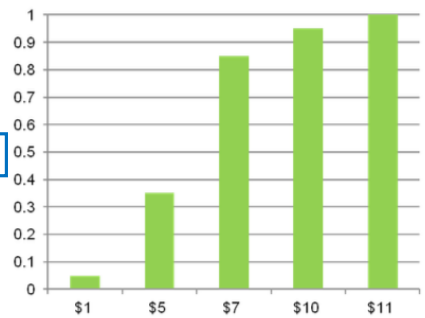
Recap: probability distribution

Dividend payout amount	Probability
$1	0.05
$5	0.30
$7	0.50
$10	0.10
$11	0.05

$$\text{pdf(Div)} = \begin{cases} P(\text{Div} = \$1) = 0.05 \\ P(\text{Div} = \$5) = 0.30 \\ P(\text{Div} = \$7) = 0.50 \\ P(\text{Div} = \$10) = 0.10 \\ P(\text{Div} = \$11) = 0.05 \end{cases}$$

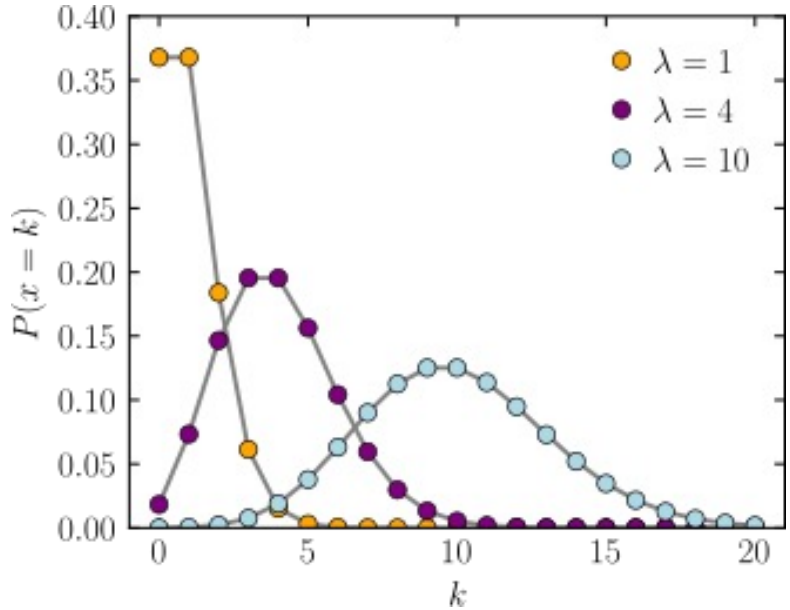


$$\text{cdf(Div)} = \begin{cases} P(\text{Div} \leq \$1) = 0.05 \\ P(\text{Div} \leq \$5) = 0.35 \\ P(\text{Div} \leq \$7) = 0.85 \\ P(\text{Div} \leq \$10) = 0.95 \\ P(\text{Div} \leq \$11) = 1.00 \end{cases}$$



Recap: probability distribution

- When X represents **the number of times an event occurs** in a fixed interval, we move to a discrete count model: the **Poisson distribution**.
- It is appropriate when we count how many times something happens, for example, the number of bankruptcies in a month, the number of network errors in one day.



Recap: probability distribution

- **PDF** of a Poisson random variable

$$P(X = x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

- Key properties

1. X is a **discrete** random variable
2. Mean: $\mu = \lambda$
3. Variance: $\sigma^2 = \lambda$
4. Standard deviation: $\sigma = \sqrt{\lambda}$

Recap: probability distribution

Complement Rule

- Instead of calculating many separate probabilities, we can subtract from 1.
- In Poisson problems, this is especially helpful for phrases like:
  - at least one event
  - at least two events
  - more than kk events

$$P(X \geq 1) = 1 - P(X = 0)$$
$$P(X \geq 2) = 1 - (P(X = 0) + P(X = 1))$$
$$P(X < 2) = P(X = 0) + P(X = 1)$$

2-4

4. Assume that the number of network errors experienced in a day on a local area network (LAN) is distributed as a Poisson random variable. The mean number of network errors experienced in a day is 2.7. Complete parts (a) through (d).

a. What is the probability that in any given day zero network errors will occur?

The probability that zero network errors will occur is _____.
(Round to four decimal places as needed.)

b. What is the probability that in any given day exactly one network error will occur?

The probability that exactly one network error will occur is _____.
(Round to four decimal places as needed.)

c. What is the probability that in any given day two or more network errors will occur?

The probability that two or more network errors will occur is _____.
(Round to four decimal places as needed.)

d. What is the probability that in any given day fewer than three network errors will occur?

The probability that fewer than three network errors will occur is _____.
(Round to four decimal places as needed.)

2-4 solution

a.

$$P(X = x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!} \qquad \mu = \lambda \qquad \sigma^2 = \lambda$$

- $P(X = 0) = \frac{e^{-2.7}(2.7)^0}{0!} = \frac{e^{-2.7}\times 1}{1} = 0.0672$

b.

- $P(X = 1) = \frac{e^{-2.7}(2.7)^1}{1!} = \frac{e^{-2.7}\times 2.7}{1} = 0.1815$

2-4 solution

c.

- $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.0672 - 0.1815 = 0.7513$

d.

- $P(X = 2) = \frac{e^{-2.7}(2.7)^2}{2!} = 0.2449$
- $P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = 0.0672 + 0.1815 + 0.2449 = 0.4936$

Recap: probability distribution

- For a continuous variable, the probability of any exact single value is **zero**. So, for continuous distributions, we are usually interested in probabilities over an interval or up to a threshold.
- One of the most important continuous probability models is the normal distribution.
- **CDF** of a normal continuous random variable

$$F(x) = P(X \leq x)$$

- A normal random variable is described by its mean and standard deviation.

$$X \sim N(\mu, \sigma^2)$$

Recap: probability distribution

- Typical questions are:
  - $P(X < a); P(a < X < b)$
  - Finding the cutoff value for a given probability
- To answer these questions efficiently, we convert X into the standard normal variable Z, using

$$Z = \frac{X - \bar{X}}{S}$$

- The Z-score tells us how many standard deviations a value is away from the mean.
- **Lookup: cumulative standardized normal table**

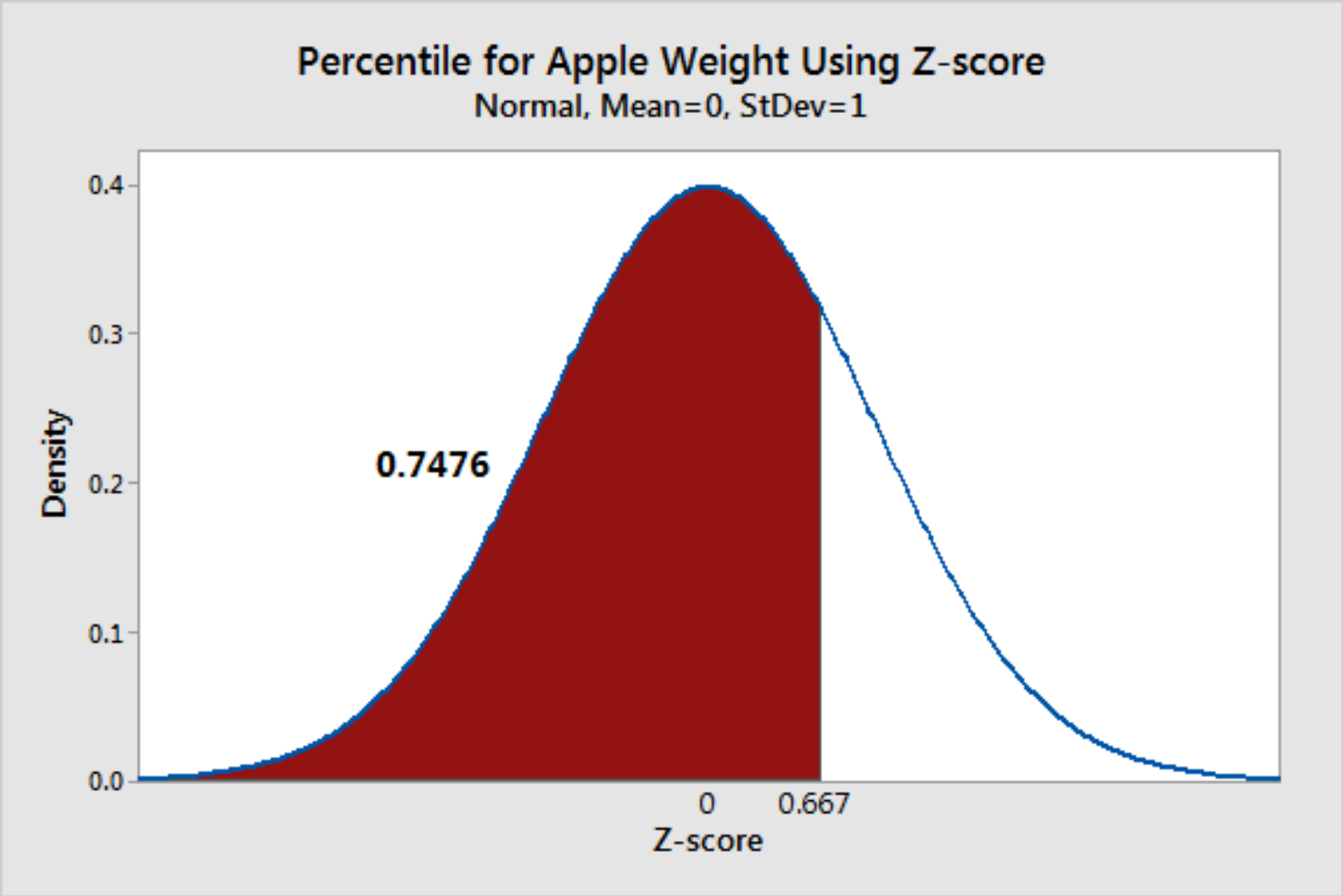
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.0	0.0014	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Negative z-table

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Positive z-table

$Standard\ Normal\ Z - table\colon\ P(Z < z)$


Percentile for Apple Weight Using Z-score
Normal, Mean=0, StDev=1
Density
0.4
0.3
0.2
0.1
0.0
0.7476
0
0.667
Z-score

Recap: probability distribution

- The table always reports a **left-tail** probability.
- **Main rules:**

1. Left tail: $P(X < a) = P\left(Z = \frac{X-\bar{X}}{s} < \frac{a-\bar{X}}{s}\right)$
2. Right tail: $P(X > a) = 1 - P(X < a) = 1 - P\left(Z = \frac{X-\bar{X}}{s} < \frac{a-\bar{X}}{s}\right)$
3. Between two values: $P(a < X < b) = P\left(Z = \frac{X-\bar{X}}{s} < \frac{b-\bar{X}}{s}\right) - P\left(Z = \frac{X-\bar{X}}{s} < \frac{a-\bar{X}}{s}\right)$

Recap: probability distribution

4. If the probability (*P*) is given first, we reverse the process:

- Find the corresponding Z-score from the table
- Covert back to the original scale using

$$Z = \frac{X - \bar{X}}{S} \rightarrow X = \bar{X} + Z \times S$$

Procedure:

- Convert X to Z
- Read the cumulative probability from the table
- Subtract from 1 or take differences when needed
- Convert back to X when the question asks for a cutoff value

2-5

5. A study found that consumers spend an average of $23 per week in cash without being aware of where it goes. Assume that the amount of cash spent without being aware of where it goes is normally distributed and that the standard deviation is $6. Complete parts (a) through (c).

Click here to view page 1 of the cumulative standardized normal distribution table.[2] Click here to view page 2 of the cumulative standardized normal distribution ta

a. What is the probability that a randomly selected person will spend more than $26?

P(X > $26) = _____ (Round to four decimal places as needed.)

b. What is the probability that a randomly selected person will spend between $8 and $22?

P($8 < X < $22) = _____ (Round to four decimal places as needed.)

c. Between what two values will the middle 95% of the amounts of cash spent fall?

The middle 95% of the amounts of cash spent will fall between X = $_____ and X = $_____.
(Round to two decimal places as needed.)

2-5 solution

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224

a.

- $X \sim N(\bar{X} = 23, \sigma = 6)$

$$Z = \frac{X - \bar{X}}{S}$$

- Q: $P(\boldsymbol{X} > 26) = P\left(\frac{X-23}{6} > \frac{26-23}{6}\right) \rightarrow P(\boldsymbol{Z} > 0.5)$?
- From the table, we know $P(Z < 0.5) = 0.6915$
- $P(Z > 0.5) = 1 - P(Z < 0.5) = 0.3085$
- $\rightarrow P(X > 26) = 0.3085$

2-5 solution

b.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.0	0.0014	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

- $X \sim N(\bar{X} = 23, \sigma = 6)$
- Q: $P(8 < X < 22) = P\left(\frac{8-23}{6} < \frac{X-23}{6} < \frac{22-23}{6}\right) \rightarrow P(-2.5 < Z < -0.17) = P(Z < -0.17) - P(Z < -2.5)$?
- Since $P(Z < -0.17) = 0.4325; P(Z < -2.5) = 0.0062$
- $P(8 < X < 22) = P(Z < -0.17) - P(Z < -2.5) = 0.4325 - 0.0062 = 0.4263$

2-5 solution

c.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

- We want the middle 95% of the distribution, which leaves 2.5% in each tail.
- From the Z-table, the critical values are $Z = \pm 1.96$

2-5 solution

c. (cont.)

- Convert back to X:

$$Z = \frac{X - \bar{X}}{S} \rightarrow X = \bar{X} + Z \times S$$

- $X = 23 \pm 1.96 \times 6$
- So the interval is:
- $11.24 < X < 34.76$
- Therefore, the middle 95% of weekly spending is between \$11.24 and \$34.76.

2-6

6. A statistical analysis of 1,000 long-distance telephone calls made by a company indicates that the length of these calls is normally distributed, with $\mu = 200$ seconds and $\sigma = 50$ seconds. Complete parts (a) through (d).
[4] Click here to view page 1 of the cumulative standardized normal distribution table.
[5] Click here to view page 2 of the cumulative standardized normal distribution table.

a. What is the probability that a call lasted less than 110 seconds?

The probability that a call lasted less than 110 seconds is _____.
(Round to four decimal places as needed.)

b. What is the probability that a call lasted between 110 and 280 seconds?

The probability that a call lasted between 110 and 280 seconds is _____.
(Round to four decimal places as needed.)

c. What is the probability that a call lasted between 50 and 110 seconds?

The probability that a call lasted between 50 and 110 seconds is _____.
(Round to four decimal places as needed.)

d. What is the length of a call if only 2.5% of all calls are shorter?

2.5% of the calls are shorter than _____ seconds.
(Round to two decimal places as needed.)

2-6 solution

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455

a.

- Let $X \sim N(200, 50)$
- We want to find: $P(X < 110)$
- Standardize: $P(X < 110) \rightarrow P\left(Z = \frac{X-200}{50} < \frac{110-200}{50}\right) < P(Z < -1.80)$
- From the Z-table: $P(Z < -1.80) = 0.0359$
- Therefore: $P(X < 110) = \mathbf{0.0359}$

2-6 solution

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706

b.

- Let $X \sim N(200, 50)$
- We want to find: $P(110 < X < 280)$
- Standardize: $P(110 < X < 280) \rightarrow P\left(\frac{110-200}{50} < Z < \frac{280-200}{50}\right) = P(-1.80 < Z < 1.60)$
- Use the Z-table: $P(-1.80 < Z < 1.60) = P(Z < 1.60) - P(Z < -1.80)$
- From the table: $0.9452 - 0.0359 = 0.9093$
- Therefore, $P(110 < X < 280) = \mathbf{0.9093}$

2-6 solution

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.0	0.0014	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036

c.

- Let $X \sim N(200, 50)$
- We want to find: $P(50 < X < 110)$
- Standardize: $P(50 < X < 110) \rightarrow P\left(\frac{50-200}{50} < Z < \frac{110-200}{50}\right) = P(-3.00 < Z < -1.80)$
- Use the Z-table: $P(-3.00 < Z < -1.80) = P(Z < -1.80) - P(Z < -3.00)$
- From the table: $0.0359 - 0.0014 = 0.0345$
- Therefore, $P(50 < X < 110) = \mathbf{0.0345}$

2-6 solution

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455

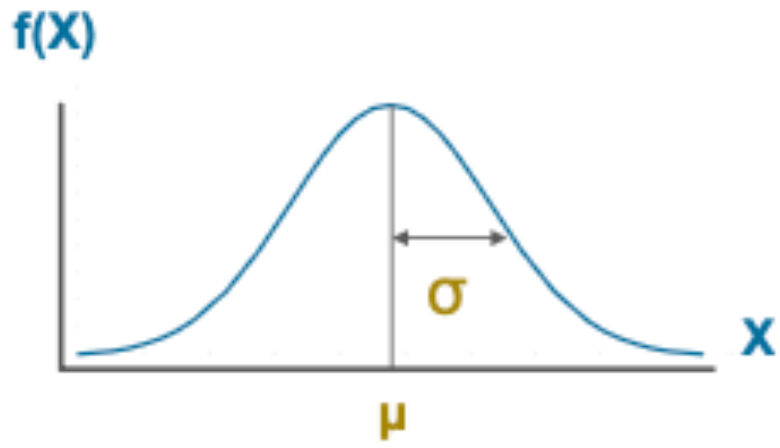
d.

- Let $X \sim N(200, 50)$
- We want the value of x such that: $P(X < x) = 0.025$
- From the Z-table, the corresponding critical value is: $Z = -1.96$
- Convert back to X: $Z = \frac{X - \bar{X}}{S} \rightarrow X = \bar{X} + Z \times S$
- $x = 200 + (-1.96) \times 50 = \mathbf{102.00}$
- Therefore, 2.5% of the calls are shorter than 102 seconds.

Recap: probability distribution

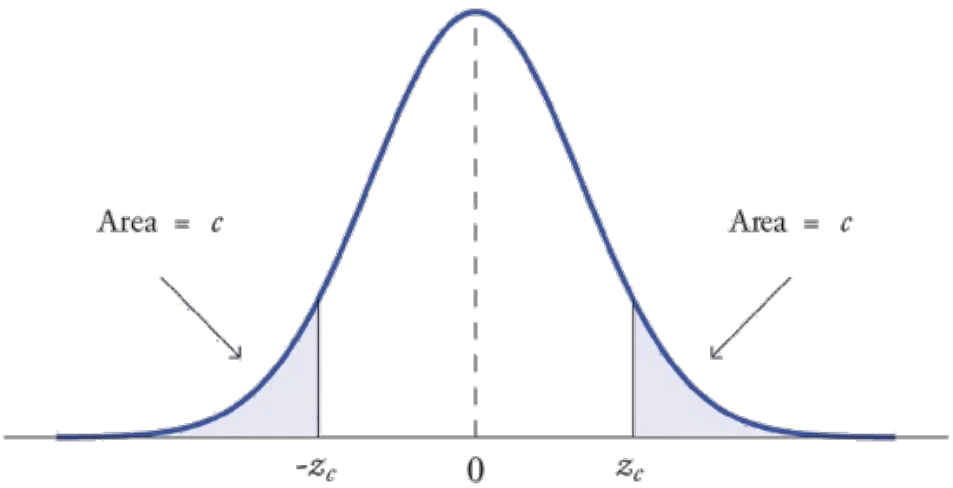
- The key properties

1. Bell shaped
2. Symmetrical
3. Mean, median, and mode are equal
4. Location is determined by the mean (μ)
5. Spread is determined by the standard deviation (σ)
6. An infinite theoretical range



Recap: probability distribution

- By the symmetry of the standard normal distribution, the left-tail and right-tail probabilities are **mirror images** around $z = 0$.
- Therefore, if there is **only one** cumulative z-table, the other side can be obtained by symmetry.
- $P(Z < -z_c) = P(Z > z_c)$
- or $P(Z > -z_c) = P(Z < z_c)$



Recap: probability distribution

- Quantitative checks for **normality**

1. Central tendency: $Mean = Median = Mode$
2. Symmetry (Skewness): $Skewness = 0$
3. Shape & Tails (Kurtosis): $Kurtosis = 0$
4. Middle data spread (The interquartile range (IQR) is the range of the middle half of a data set, representing the spread between the 25th and 75th percentile): $IQR = 1.33 \times Standard\ Deviation$
5. Total data spread: $Range = 6 \times Standard\ Deviation$

Recap: probability distribution

- The Boxplot: A Graphical display of the data based on the five-number summary:

Recap: probability distribution

Excel Formulas for Normality

- $Mean = AVERAGE()$
- $Median\ (Q2) = MEDIAN()$
- $Mode = MODE()$
- $Standard\ deviation = STDEV()$
- $Kurtosis = KURT()$
- $Skewness = SKEW()$
- $Q1, Q3, IQR = QUARTILE()$
- $Range = MAX() - MIN()$

2-8

8. One common method for measuring the size of a company is to use its market capitalization, which is computed by multiplying the number of stock shares by the price of a share of stock. The data set below contains the market capitalization values, in dollars, for 30 companies. Complete parts (a) through (c).

44.7	62.2	45.0	20.7	126.0	145.6	28.5	91.2	83.0	75.7
255.7	52.4	314.1	38.6	29.6	107.4	22.6	21.0	176.9	46.1
181.1	122.3	209.7	251.8	62.5	83.1	193.5	284.6	121.1	22.0

Click here to view page 1 of the cumulative standardized normal distribution table.[6]
Click here to view page 2 of the cumulative standardized normal distribution table.[7]

a. Decide whether the market capitalization of these companies appears to be approximately normally distributed by comparing data characteristics to theoretical properties.

Based on the theoretical properties of the normal distribution, do the data appear to be approximately normally distributed?

- ○ No
- ○ Yes

2-8 solution

- No
- For this set, the mean figure is higher than the median, with a negative Kurtosis of -0.1564 and Skewness of 0.9375, the interquartile range (IQR) is higher than 1.33 times that of standard deviation, the range is less than 6 times the standard deviation, thus the distribution is not normal.

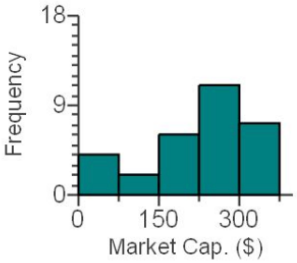
Please refer to the Excel solution for detailed steps.

2-8 (cont.)

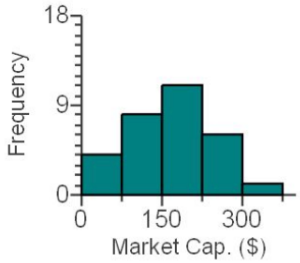
c. Decide whether the market capitalization of these companies appears to be approximately normally distributed by constructing a histogram.

Construct a histogram. Choose the correct histogram below.

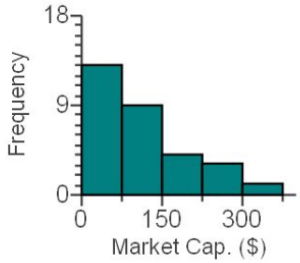
○ **A.**



○ **B.**



○ **C.**



What is the best description of the given data based on the histogram?

○ **A.** The data are skewed to the right.
○ **B.** The data are skewed to the left.
○ **C.** The data appear to be normally distributed.

2-8 solution (cont.)

- C
- A

Please refer to the Excel solution for detailed steps.

2-9

9. The data set below contains the yields for money market accounts at 21 banks in 2007. Complete parts (a) and (b).

0.76	1.07	0.68	0.44	1.87	2.22	0.51	1.59	1.32	1.02	3.78
0.79	4.55	2.96	0.63	0.48	1.69	0.65	0.49	0.45	2.46	

Click here to view page 1 of the cumulative standardized normal distribution table.[8]
Click here to view page 2 of the cumulative standardized normal distribution table.[9]

a. Decide whether the data appear to be approximately normally distributed by comparing data characteristics to theoretical properties.

Based on the theoretical properties of the normal distribution, do the data appear to be approximately normally distributed?

- ○ No
- ○ Yes

2-9 solution

- No
- For this set, the mean figure is higher than the median, with a Kurtosis of 1.4522 and Skewness of 1.4243, the interquartile range is lower than 1.33 times the standard deviation, the range is less than 6 times the standard deviation, thus the distribution is not normal.

Please refer to the Excel solution for detailed steps.

2-10

10. The data table shows the yields for a money market account, a one-year certificate of deposit (CD), and a five-year CD for 13 banks. Complete parts (a) through (c).

[1] Click the icon to view the data table.

a. List the five-number summary for the yields of a money market account, a one-year CD, and a five-year CD.

The five-number summary for the yields of a money market account is _____, _____, _____, _____, _____
(Use ascending order. Do not round.)

The five-number summary for the yields of a one-year CD is _____, _____, _____, _____, _____.
(Use ascending order. Do not round.)

The five-number summary for the yields of a five-year CD is _____, _____, _____, _____, _____.
(Use ascending order. Do not round.)

1: Data Table

Money Market		One-Year CD		Five-Year CD	
1.26	0.57	3.66	3.44	3.52	4.43
1.37	1.98	3.97	2.81	4.35	4.71
1.21	0.56	4.26	3.41	3.79	3.54
0.76	1.06	2.95	3.76	4.68	3.75
1.29	0.41	4.52	3.81	4.83	3.75
0.96	1.31	4.72	3.51	4.92	4.61
0.98		2.98		4.26	

2-10 solution

a.

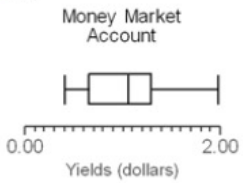
- Money Market: 0.41 0.76 1.06 1.29 1.98
- One-Year CD: 2.81 3.41 3.66 3.97 4.72
- Five-Year CD: 3.52 3.75 4.35 4.68 4.92

2-10 (cont.)

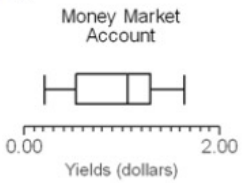
b. Construct a boxplot for the yields of the money market account, one-year CD, and a five-year CD.

Which of the following is the boxplot for the yields of a money market account?

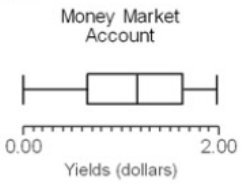
A.



B.

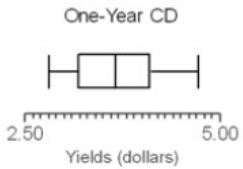


C.

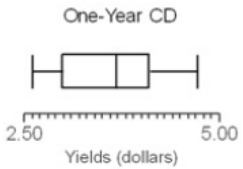


Which of the following is the boxplot for the yields of a one-year CD?

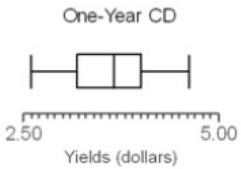
A.



B.

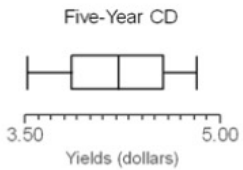


C.

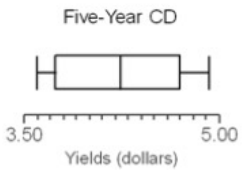


Which of the following is the boxplot for the yields of a five-year CD?

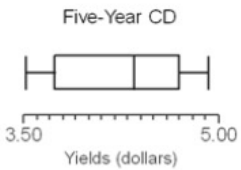
A.



B.

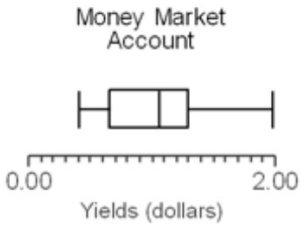


C.

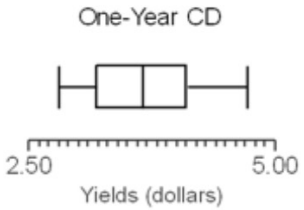


2-10 solution (cont.)

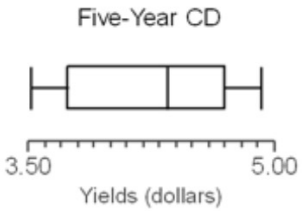
A.



A.



C.



2-10 (cont.)

c. What similarities and differences are there in the distributions for the yields of a money market account, a one-year CD, and a five-year CD?

- ○ **A.** The three kinds of investments all tend to have the same yield.
- ○ **B.** There does not tend to be much variation in the yields of a money market account whereas the yields of the other kinds of investments vary widely.
- ○ **C.** A five-year CD tends to have a higher yield than either of the other kinds of investments.
- ○ **D.** The yields of a one-year CD are symmetrically distributed whereas the yields of the other kinds of investments are both skewed.

- C. A five-year CD tends to have a higher yield than either of the other kinds of investments.

Recap: Uniform Distribution

- Another important probability distribution is the uniform distribution.
- Core idea: **all possible outcomes are equally likely.**
- If

$$X \sim U(a, b)$$

- Then all values between a and b are equally likely.
- PDF

$$f(x) = \frac{1}{b-a} \qquad a \leq x \leq b$$

And $f(x) = 0$ otherwise

Recap: Uniform Distribution

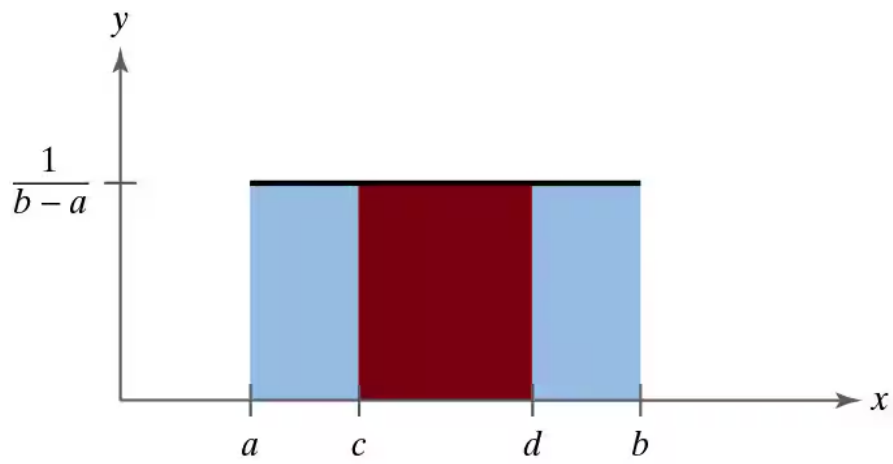
- CDF
- For any $a \leq c \leq d \leq b$,

$$P(c \leq X \leq d) = \frac{d-c}{b-a}$$

- Mean and standard deviation

$$\mu = \frac{a+b}{2}$$

$$\sigma = \frac{b-a}{\sqrt{12}}$$



2-7

7. A study of the time spent shopping in a supermarket for 20 specific items showed an approximately uniform distribution between 15 minutes and 25 minutes. **a.** What is the probability that the shopping time will be between 17 and 22 minutes? **b.** What is the probability that the shopping time will be less than 21 minutes? **c.** What are the mean and standard deviation of the shopping time?

a. The probability that the shopping time will be between 17 and 22 minutes is _____.
(Type an integer or a decimal.)

b. The probability that the shopping time will be less than 21 minutes is _____.
(Type an integer or a decimal.)

c. The mean of the given uniform distribution is $\mu =$ _____.
(Type an integer or a decimal.)

The standard deviation of the given uniform distribution is $\sigma =$ _____.
(Round to four decimal places as needed.)

2-7 solution

- Let $X \sim U(15, 25)$
- The distribution is uniform from 15 to 25 minutes
- For any $a \leq c \leq d \leq b$,

$$P(c \leq X \leq d) = \frac{d-c}{b-a}$$

- $P(17 < X < 22) = \frac{22-17}{25-15} = 0.5$
- $P(X < 21) = \frac{21-15}{25-15} = 0.6$
- $\mu = \frac{a+b}{2} = \frac{15+25}{2} = 20$ $\qquad \sigma = \frac{b-a}{\sqrt{12}} = \frac{25-15}{\sqrt{12}} = 2.8868$